

Building the first reliable model for on-device agents

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Access slides



On-device agent demo

Try it out yourself! Setup instructions to power your local agent with LFM2.5-2.6B:

<https://docs.liquid.ai/examples/agent-harnesses>

SPEED

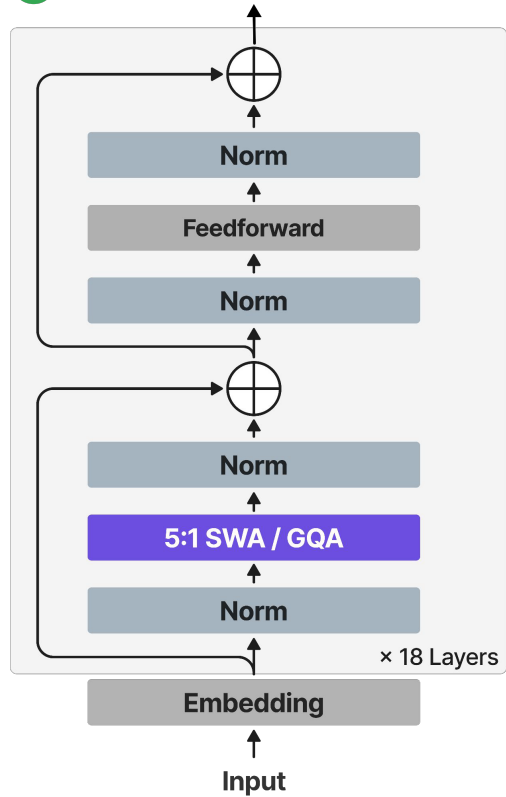
MEMORY

QUALITY

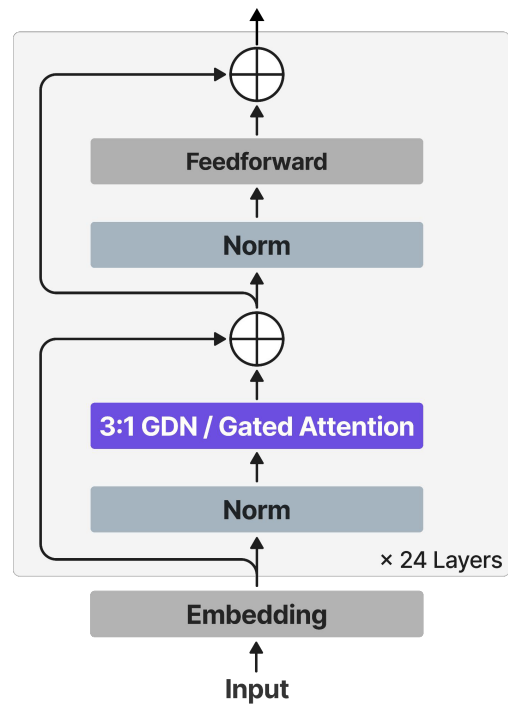
How to make it small and fast

(Architecture)

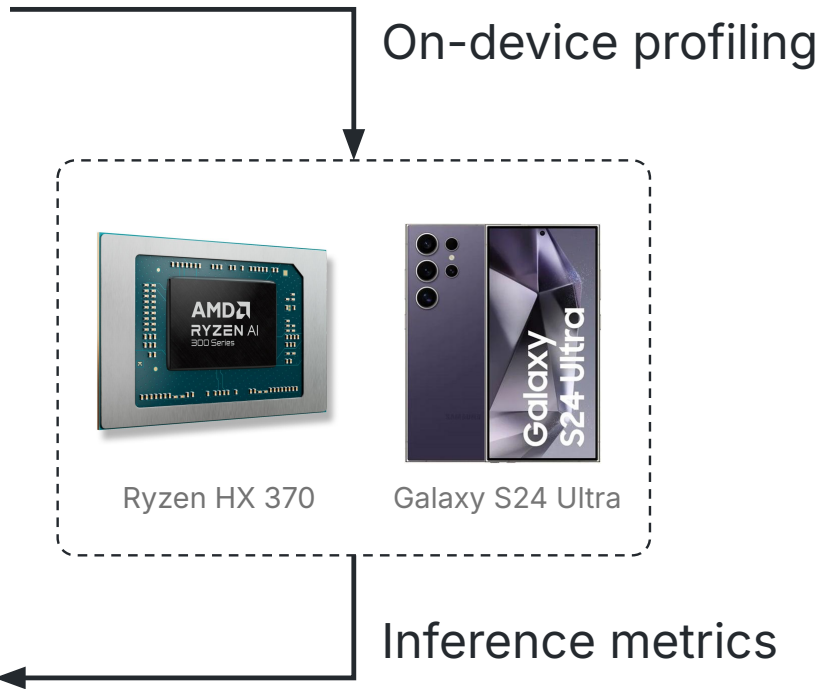
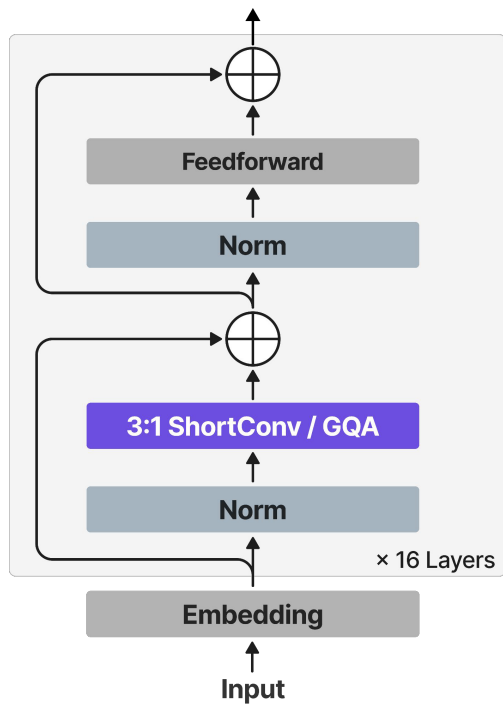
 Gemma 3 270M [LLM]



 Qwen3.5-0.8B [VLM]

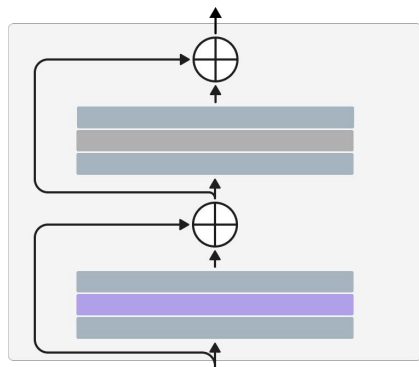


LFM2.5-350M



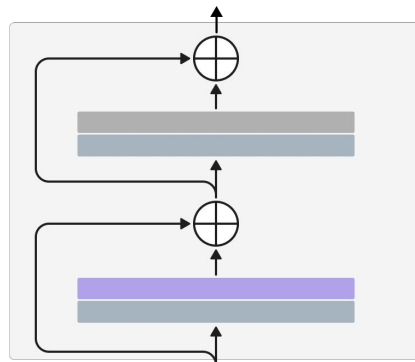
Effective size ~100M

 Gemma 3 270M [LLM]



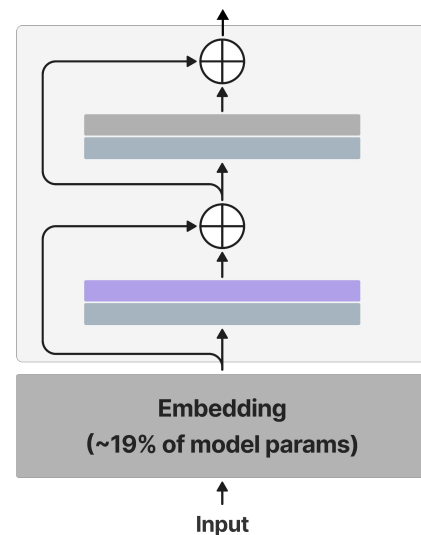
Effective size ~570M

 Qwen3.5-0.8B [VLM]



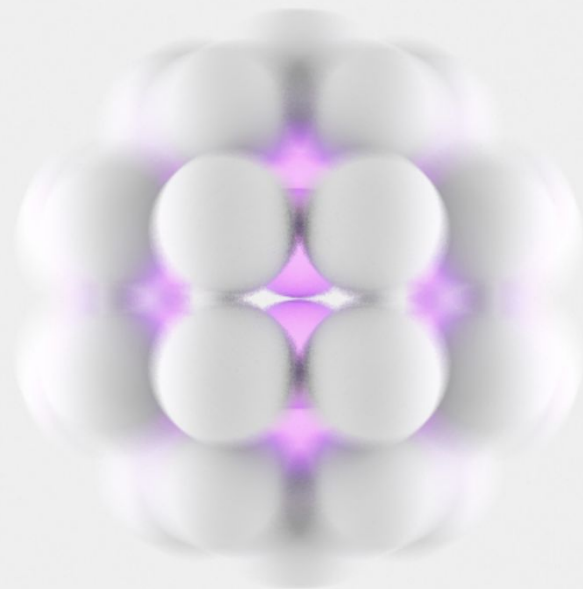
Effective size ~290M

 LFM2.5-350M



How to make it good at agentic tasks

(Training)



State of modern post-training

4.1 Multi-Teacher On-Policy Distillation (MOPD): A New Post-Training Paradigm

Modern language models increasingly rely on extensive post-training to enhance their intelligence and capabilities. However, current post-training pipelines face fundamental challenges: *capability imbalance*, where improving one skill causes regressions in others (the “see-saw” effect), and *learning inefficiency*, where existing approaches fail to fully leverage training signals when combining knowledge from multiple specialized models.

MiMo-V2-Flash Tech Report (Jan 2026)

5.1.2. On-Policy Distillation

After training multiple domain-specific experts via specialized fine-tuning and reinforcement learning, we employ [multi-teacher On-Policy Distillation](#) (OPD; Gu et al. 2024; Lu and Lab 2025) as the primary technique for merging expert capabilities into the final model. OPD has emerged as an effective post-training paradigm for efficiently transferring the knowledge and capabilities of domain experts to a single, unified model. This is achieved by having the student learn from the output distributions of teacher models on its own generated trajectories. Formally, given a set of N expert models $\{\pi_{E_1}, \pi_{E_2}, \dots, \pi_{E_N}\}$, the OPD objective function is defined as:

DeepSeek V4 Tech Report (May 2026)

4.4. Multi-domain On-Policy Distillation (MOPD)

While well-designed Cascade RL substantially reduces catastrophic forgetting compared with vanilla sequential RL in an arbitrary order, it does not fully eliminate capability drift as the number of training environments increases. In practice, we observe noticeable fluctuations across different benchmark categories tracked throughout training, and the dominant trade-offs differ by stage. For example, certain RLVR training often reduces model entropy and shortens reasoning traces, thus can negatively impact mathematical reasoning performance, while RLHF-oriented optimization can partially trade off against instruction-following behavior. These observations motivate an additional training stage for re-balancing capabilities within the Cascade RL process.

Nemotron-Cascade 2 Tech Report (Mar 2026)

3.3 Agentic RL

To facilitate agentic performance of GLM-5, we develop a fully asynchronous and decoupled RL framework and optimize GLM-5 in coding and search agent tasks. Naive synchronous RL suffers from severe GPU idle time during long-horizon agent rollouts. By decoupling inference and training engines via a central Multi-Task Rollout Orchestrator, we achieve high-throughput joint training across diverse agentic workloads.

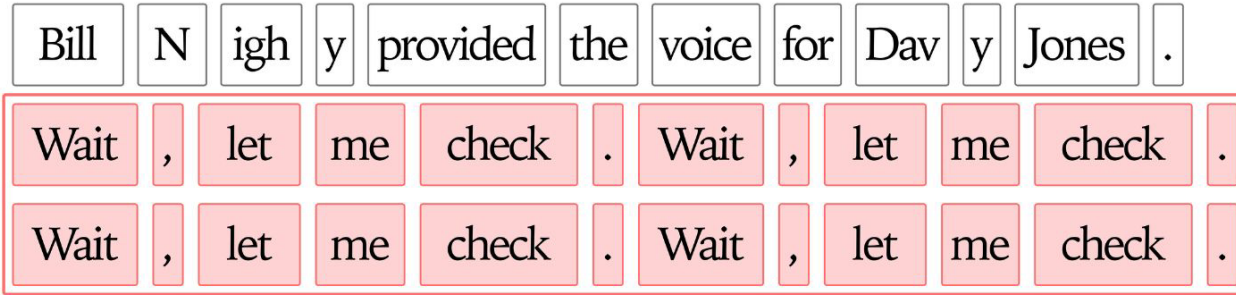
GLM-5 Tech Report (Feb 2026)

5.3 Infra for 1M Agentic RL

Scaling agentic RL for a model as large as Kimi K3 to million-token contexts under a bounded compute budget makes resource efficiency a first-order goal. We therefore develop long-context RL infrastructure for efficient training and rollout, together with high-performance, resumable sandboxes for long-horizon environment interaction.

Kimi K3 Tech Report (Aug 2026)

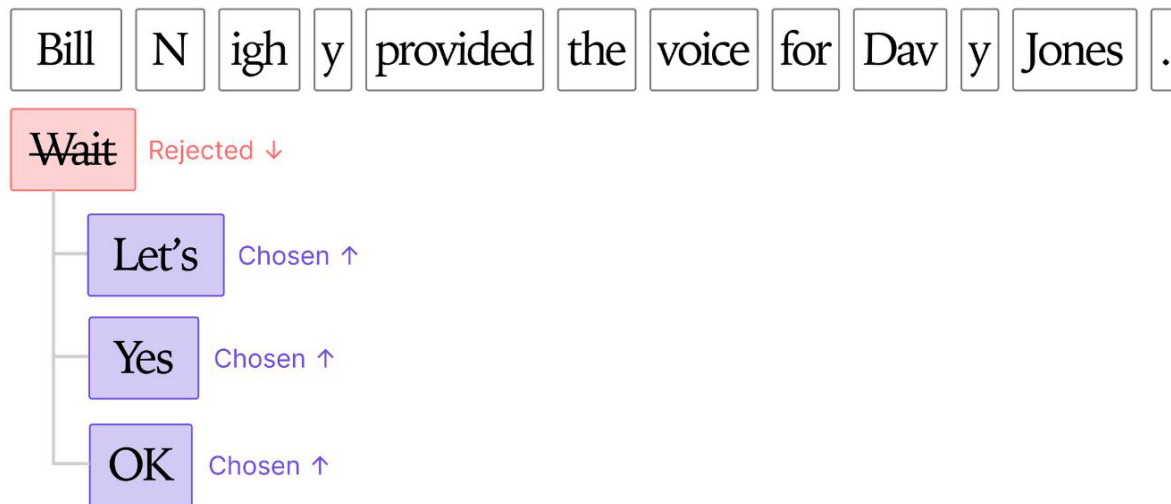
Who voiced Davy Jones in the Pirates of the Caribbean series?



Doom Loop

Antidoom with Final Token Preference Optimization (FTPO)

Who voiced Davy Jones in the Pirates of the Caribbean series?



Reinforcement Learning

What is $4 + (8 \div 2)$?

Trajectory

4 + 8 is 12 , and 12 $\div$ 2 is 6 .

Ground truth

8

reward = 0

On-Policy Distillation

What is $4 + (8 \div 2)$?

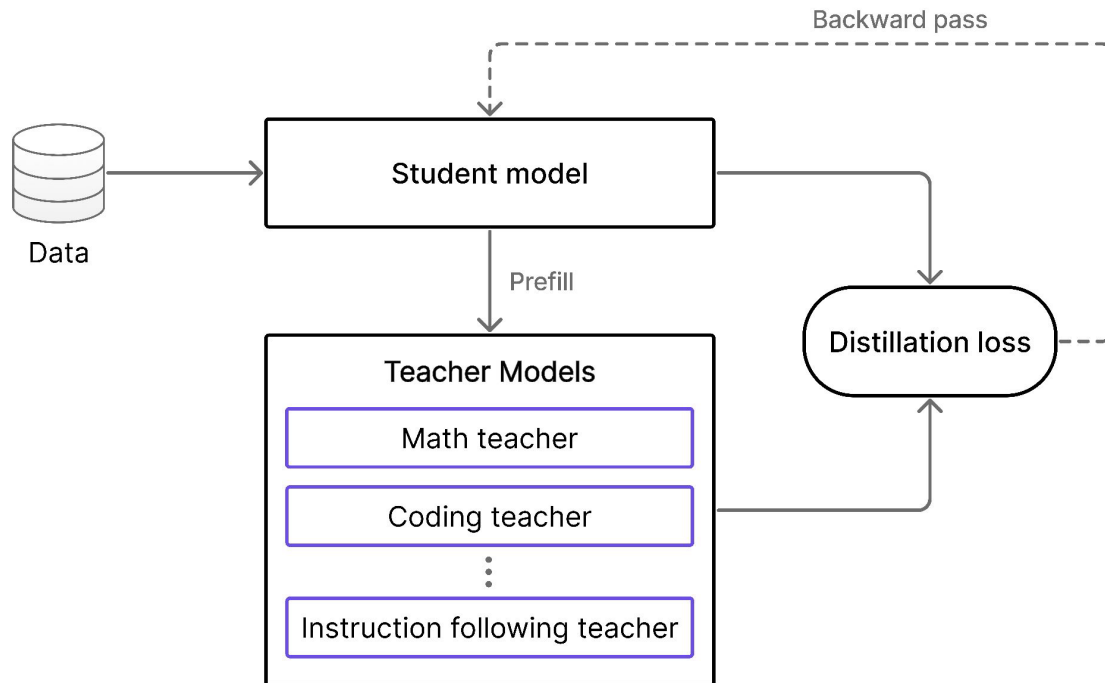
Student trajectory



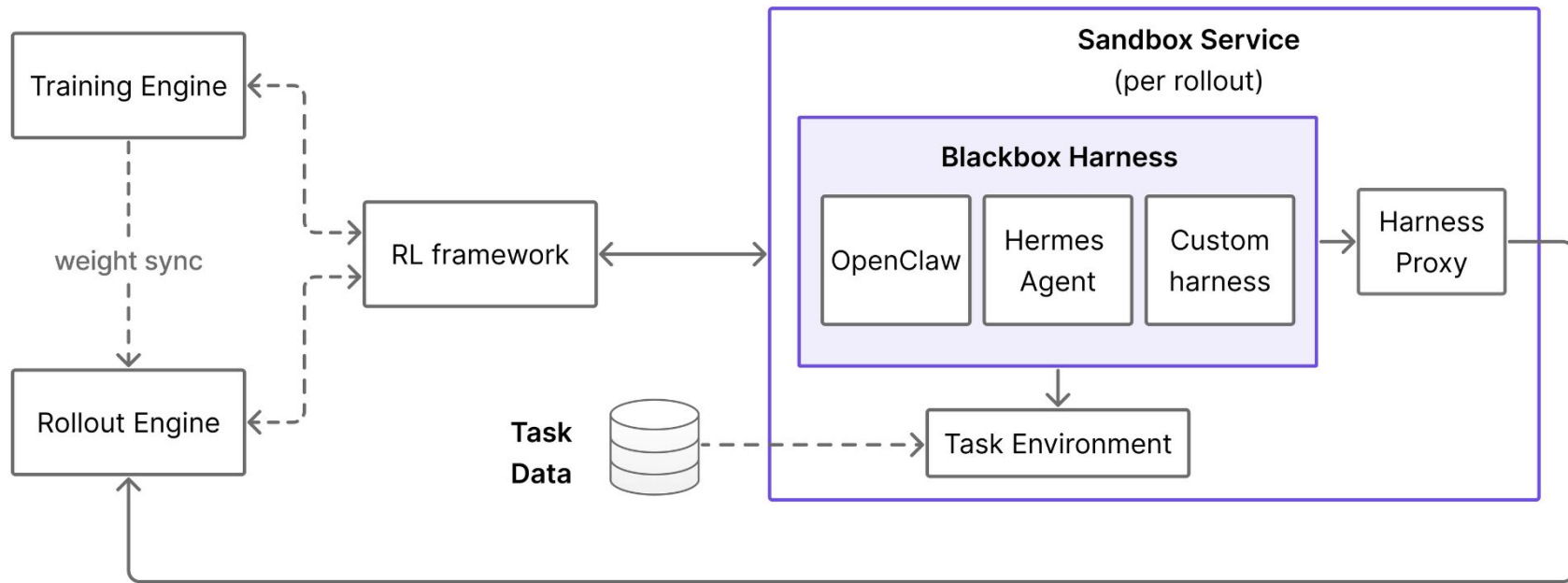
Teacher's
per-token probability

dense signal

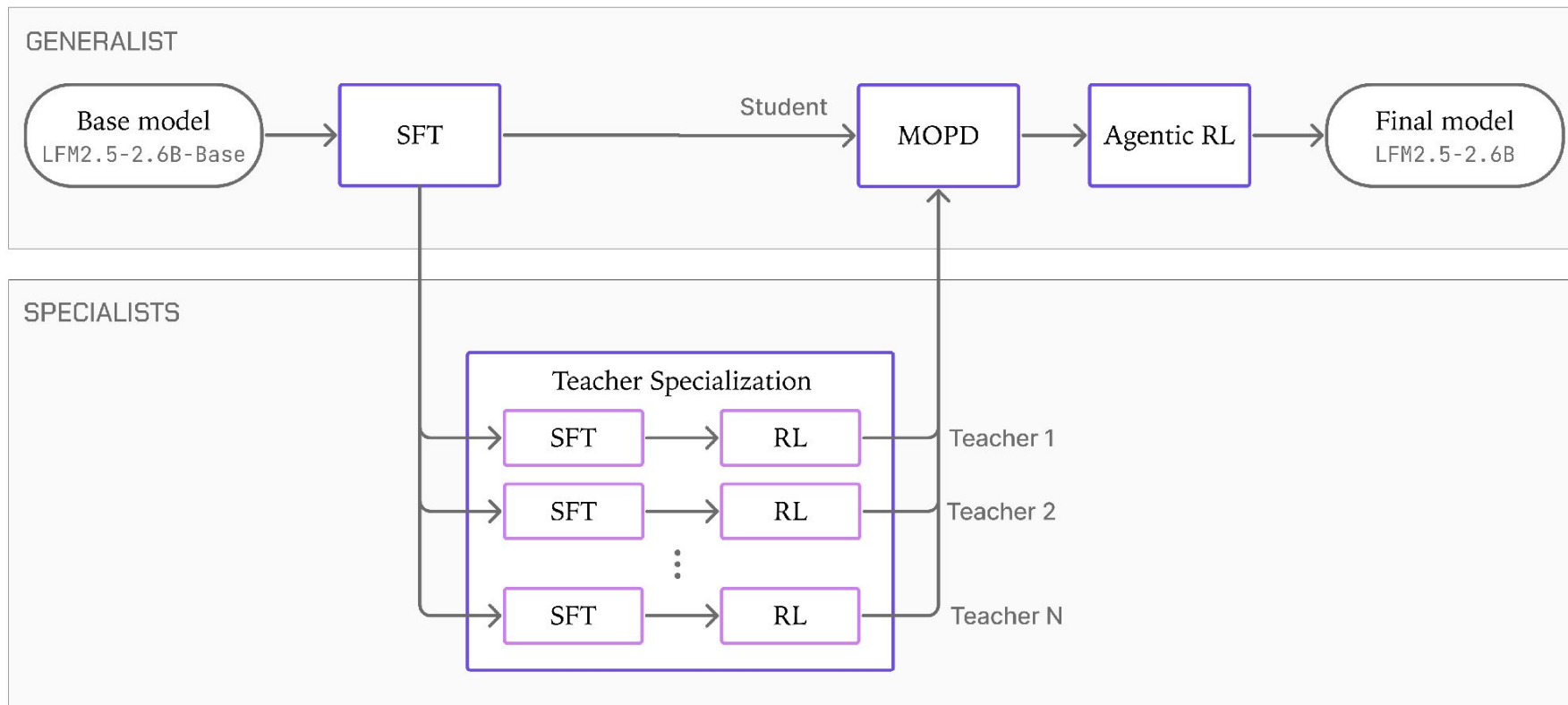
Multi-Domain On-Policy Distillation (MOPD)



Agentic Reinforcement Learning



Post-training recipe for LFM2.5-2.6B



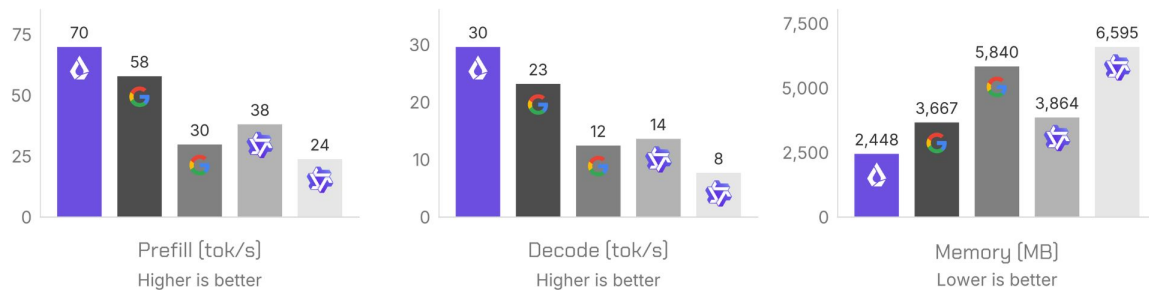
Small and fast on-device
via architecture.

LFM2.5-2.6B Gemma-4-E2B Gemma-4-E4B Qwen3.5-4B Qwen3.5-9B

On-device inference

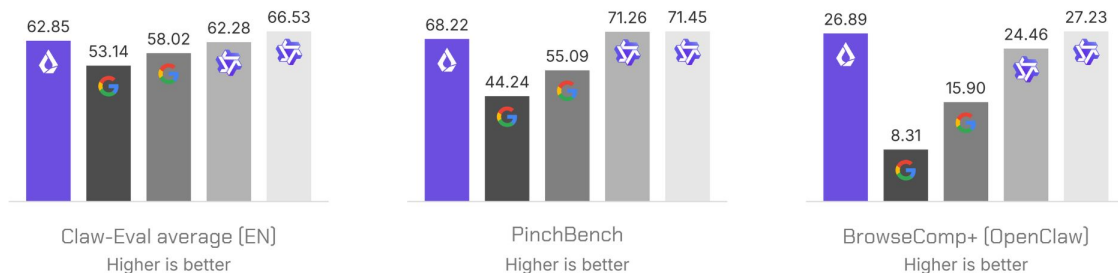
Qualcomm Snapdragon (Galaxy SM-S948U1)

Q4_K_M quantization · 4K-token prefill / memory · 2K-token decode



Performance evaluations for agentic benchmarks

Good at agentic tasks
via training.



deploy agents everywhere.

deploy agents everywhere.

deploy agents everywhere.

deploy agents everywhere.

deploy agents everywhere.

deploy agents everywhere.

Thank you.

Let's build something on-device!

<https://huggingface.co/LiquidAI/LFM2.5-2.6B>

X @helloiamleonie

in leonimonigatti

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